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HydroSmart-MA: An open-source precision irrigation system using Penman-Monteith and real-time Sentinel-2 satellite data

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Abstract

Morocco is experiencing an unprecedented water stress crisis, particularly in critical agricultural regions like Souss-Massa. Traditional irrigation methods lead to massive water waste, while static estimation models fail to capture micro-climate and crop growth variations. To address this, we present HydroSmart-MA, an open-source, multi-platform software suite designed for precision irrigation scheduling. The platform consists of a Next.js administrative dashboard and a Flutter mobile application for farmers, backed by a unified Supabase cloud database. HydroSmart-MA calculates daily reference evapotranspiration (ET0) in real-time using the FAO-56 Penman-Monteith equation driven by local weather API inputs. Critically, the system incorporates real-time Sentinel-2 satellite imagery via the Agro Monitoring API to dynamically calculate crop coefficients (Kc) from Normalized Difference Vegetation Index (NDVI) measurements, and to model daily soil water depletion. This paper details the software architecture, the underlying physical models, and demonstrates its potential to reduce agricultural water consumption by 15-20%.

Keywords: Precision Agriculture, Evapotranspiration, Remote Sensing, Next.js, Flutter, Souss-Massa

Code metadata

Current code version	v1.0.0
Permanent link to code/repository	https://github.com/aadmin/HydroSmart
Legal code license	MIT License
Code versioning system used	git
Languages, tools and services used	TypeScript, Dart, SQL (Next.js, Flutter, Supabase)
Supported Operating Systems	Windows, macOS, Linux, Android, iOS

1. Motivation and significance

Morocco's agricultural sector represents a cornerstone of the national economy but consumes over 80% of the country's freshwater resources. With climate change accelerating aridification, major agricultural basins like Souss-Massa are depleting their aquifers at alarming rates. Precision irrigation is no longer an optional optimization but a vital necessity for agricultural sustainability. Existing proprietary smart irrigation software often relies on expensive in-situ IoT sensors (soil moisture probes, local weather stations) that are financially inaccessible to smallholder Moroccan farmers. To bridge this gap, HydroSmart-MA leverages free satellite data and public weather APIs to provide high-precision irrigation recommendations without requiring any hardware installation on the farm.

2. Software description

HydroSmart-MA is designed with a service-oriented, multi-platform architecture sharing a single PostgreSQL relational database hosted on Supabase. The administrative web dashboard is built using Next.js 16, React 19, and Tailwind CSS. It provides agricultural administrators and researchers with tools to monitor region-wide water usage, manage crop profiles, view historical irrigation charts using Recharts, and verify recommendation quality.

The farmer mobile application is built using Flutter. It provides smallholders with an intuitive interface to register their fields (plots), pick their crops, and pinpoint their location on an interactive map. It communicates with the backend via HTTP API endpoints to fetch recommendations and log irrigation history.

3. Core mathematical models

The software engine centralizes three primary agro-meteorological algorithms. First, daily reference evapotranspiration (ET0) is computed in real-time using the FAO-56 Penman-Monteith equation, utilizing atmospheric data fetched from OpenWeather. Second, rather than utilizing static tables, HydroSmart-MA computes the crop coefficient (Kc) dynamically using Sentinel-2 NDVI values obtained via the Agro Monitoring API: Kc = 1.25 * NDVI + 0.2 (bounded between 0.15 and 1.25). Third, when satellite-radar soil moisture (SM) is available, the net irrigation requirement is determined by calculating the root zone depletion Dr = 1000 * (FC - SM) * Zr, where FC is the soil Field Capacity and Zr is the crop-specific root depth. Irrigation is triggered only when depletion exceeds the critical Readily Available Water (RAW) threshold.

4. Illustrative example

Consider a 1.2-hectare loamy tomato field located in Souss-Massa, Morocco (Field Capacity FC = 0.27, Permanent Wilting Point PWP = 0.12, Root Depth Zr = 0.6m, depletion fraction p = 0.4). On a hot summer day, the weather API returns parameters yielding ET0 = 5.2 mm/day. The satellite returns an NDVI of 0.68, leading to a resolved Kc = 1.05 and crop evapotranspiration ETc = 5.46 mm/day. The satellite radar estimates the soil moisture at SM = 0.18. The system computes: TAW = 1000 * (0.27 - 0.12) * 0.6 = 90.0 mm; RAW = 0.4 * TAW = 36.0 mm; and current depletion Dr = 1000 * (0.27 - 0.18) * 0.6 = 54.0 mm. Since depletion (54.0 mm) exceeds RAW (36.0 mm), irrigation is triggered. With a drip efficiency of 0.85, the gross irrigation depth required is 54.0 / 0.85 = 63.5 mm. For 1.2 hectares, the platform recommends a volume of 762 m3 to be applied.

5. Impact and conclusions

HydroSmart-MA offers an accessible, zero-hardware solution to precision agriculture in water-scarce regions. By using satellite indices and weather inputs to model exact crop water requirements, preliminary pilot simulations indicate water savings of 15-20% compared to traditional calendar-based irrigation practices. Future work will focus on integrating machine learning algorithms to predict crop water needs 3 days in advance.